

👉 XAI Lecture 10

Sanity Checks for Saliency Maps & OpenXAI

FAMAF - UNC

First Semester 2026



eXplainable
Artificial Intelligence

Disclaimer ---

This course is based on

Explainable Artificial Intelligence

From Simple Predictors to Complex Generative Models

Spring 2023, Harvard University

<https://interpretable-ml-class.github.io/>

6 Nov 2020

Sanity Checks for Saliency Maps

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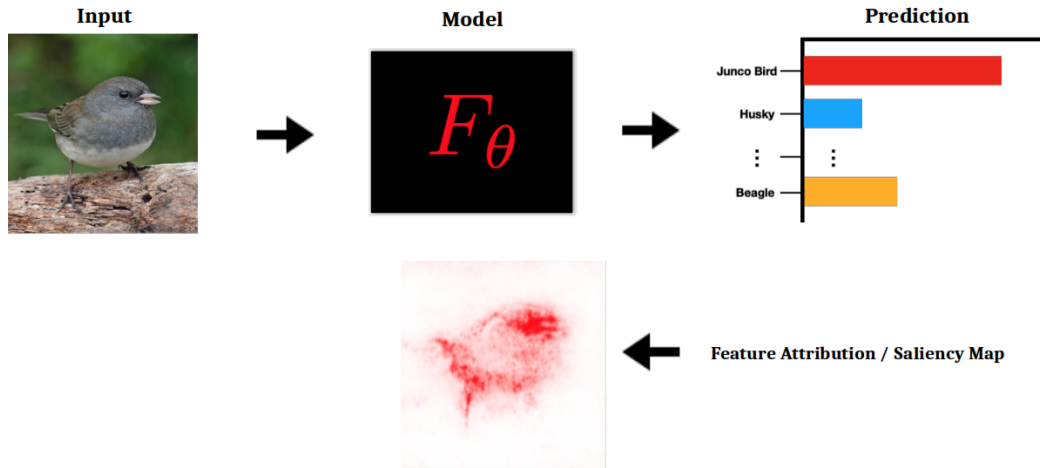
Abstract

Saliency methods have emerged as a popular tool to highlight features in an input deemed relevant for the prediction of a learned model. Several saliency methods have been proposed, often guided by visual appeal on image data. In this work, we propose an actionable methodology to evaluate what kinds of explanations a given method can and cannot provide. We find that salience scales on visual assessment

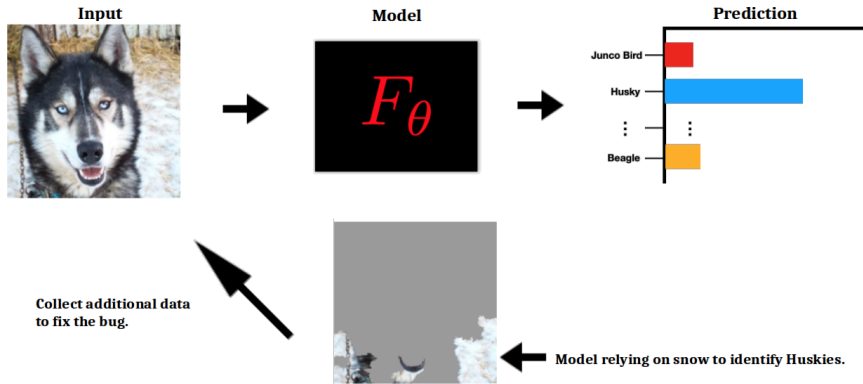
(Adebayo et al. 2018)

🍷 Feature Attributions / Saliency Maps ---

What parts of the input are 'most important' for the model prediction?



🍷 Identifying Shortcuts ---



A **shortcut** is a specific type of bias — the model learns a spurious correlation that works well on training data but fails to generalize. Instead of learning the relevant feature (the dog), it learns an easier, correlated signal (the snow).

🍷 Feature Attribution / Saliency Maps ---

Feature attribution method: assigns an output 'relevance' score to each dimension of the input.

$$F : \mathbb{R}^d \rightarrow \mathbb{R}^c \quad \text{Model}$$

$$F_i : \mathbb{R}^d \rightarrow \mathbb{R} \quad \text{class specific logit}$$

An attribution method produces a map $E : \mathbb{R}^d \rightarrow \mathbb{R}^d$ of the same shape as the input, where each value $E_j(x)$ indicates how much dimension j contributed to the prediction of class i .

🍷 Input-Gradient / Saliency / Gradient ---

Input-Gradient:

$$\nabla_x F_i(x) \in \mathbb{R}^d$$

- Same dimension as the input
- Gradient w.r.t. **Input** for class **Logit** i

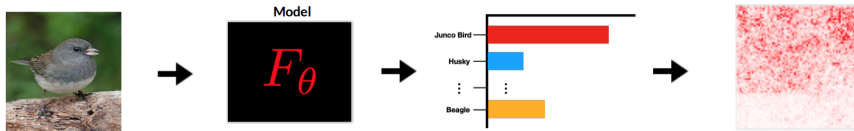
Intuitively, $\nabla_x F_i(x)$ answers:

"if I perturb input dimension j slightly, how much does the predicted score for class i change?"

Large values indicate dimensions the model is most sensitive to near x .

(Baehrens et al. 2010; Simonyan, Vedaldi, and Zisserman 2014)

🍷 Input-Gradient / Saliency / Gradient - Demo ---



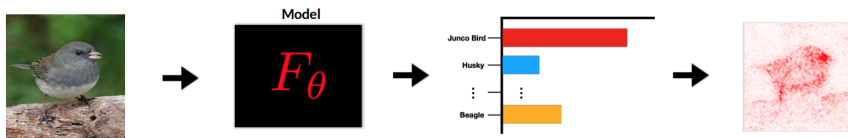
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🍷 Integrated Gradients ---

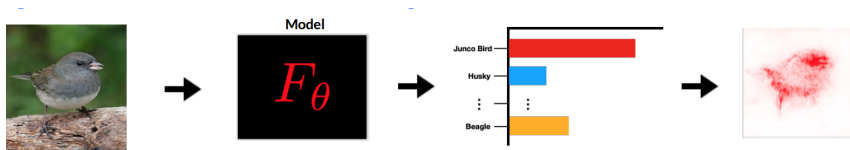


$$(x - \tilde{x}) \times \int_{\alpha=0}^1 \frac{\partial F(\tilde{x} + \alpha \times (x - \tilde{x}))}{\partial x}$$

Path integral: 'sum' of interpolated gradients from **Baseline input** $\tilde{x}$ to x .

(Sundararajan, Taly, and Yan 2017)

🍷 SmoothGrad ---



$$\frac{1}{N} \sum_i^N \nabla_{(x+\epsilon)} F_i(x + \epsilon)$$

Average of gradients computed on input x perturbed with **Gaussian noise** ϵ .

(Smilkov et al. 2017)

🍷 Guided Backprop: "Modified Backprop" ---



activation:

$$f_i^{l+1} = \text{relu}(f_i^l) = \max(f_i^l, 0)$$

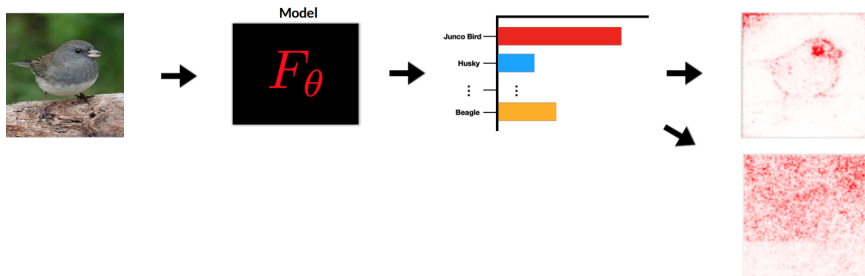
backpropagation:

$$R_i^l = (f_i^l > 0) \cdot R_i^{l+1}, \quad R_i^{l+1} = \frac{\partial f^{out}}{\partial f_i^{l+1}} R_i^l = (f_i^l > 0) \cdot \boxed{(R_i^{l+1} > 0)} \cdot R_i^{l+1}$$

guided backpropagation:

Additional gate: only backpropagate **positive** relevance signals

🍷 Are Saliency Maps Reliable? ---



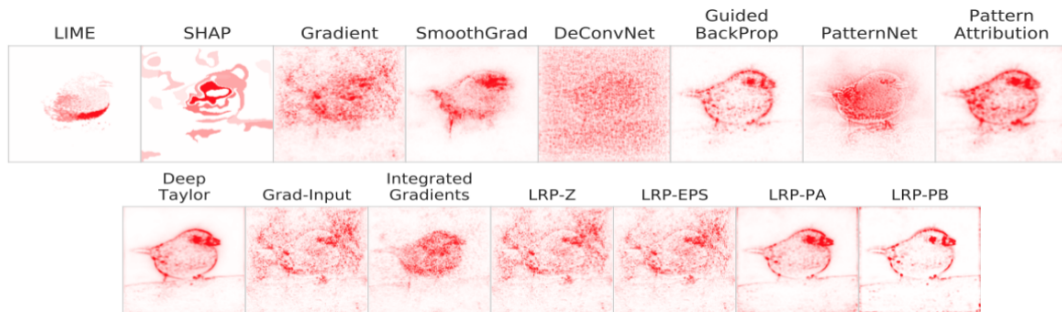
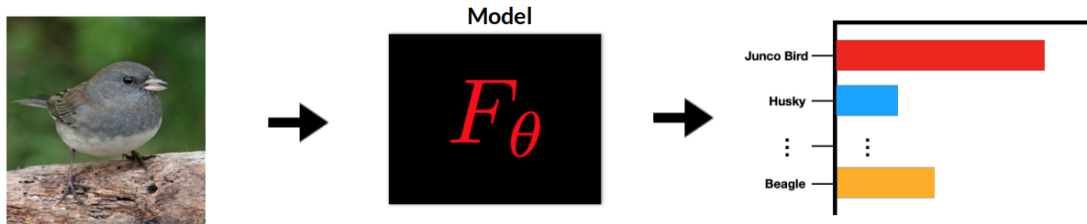
Both maps are produced by the **same method** on the **same input** — but one comes from a trained model and the other from a **randomly initialized** network.

- **Left:** explanation from a **trained** model.
- **Right:** explanation from a **randomly initialized** model.

Can you tell which is which?

Visual inspection alone is not a reliable criterion for evaluating saliency methods.

🍷 Recap: Many Saliency Methods ---



Recap: Additional Methods ---

- **Class Activation Mapping** (Zhou et al. 2016)
- **Meaningful Perturbation** (Fong et al. 2017)
- **RISE** (Petsuik et al. 2018)
- **Extremal Perturbations** (Fong & Patrick 2019)
- **DeepLift** (Shrikumar et al. 2018)
- **Expected Gradients** (Erion et al. 2019)
- **GradCAM / Guided GradCAM** (Selvaraju et al. 2016)
- **Occlusion** (Zeiler et al. 2014)
- **Prediction Difference Analysis** (Gu et al. 2019)
- **Internal Influence** (Leino et al. 2018)

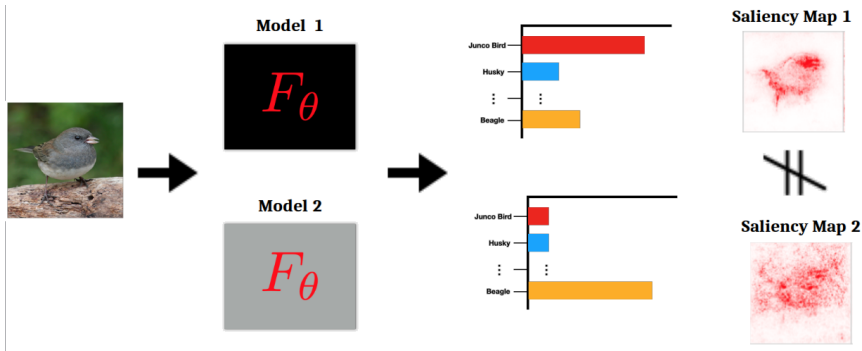
See for additional methods: Samek & Montavon et al. 2020

'Sanity Checks' ---

Intuitive 'principles' that an attribution method should satisfy.

- **'Model' Faithfulness:** is the 'explanation' sensitive to model parameters?
 - ▶ Test: change the model weights and measure corresponding change in explanation.
 - ▶ Operationalize by reinitialization of model weights.
- **Data Faithfulness:** is the attribution sensitive to training data?
 - ▶ Test: change training label and measure corresponding change in explanation.
 - ▶ Operationalize by randomization labelling in training data.

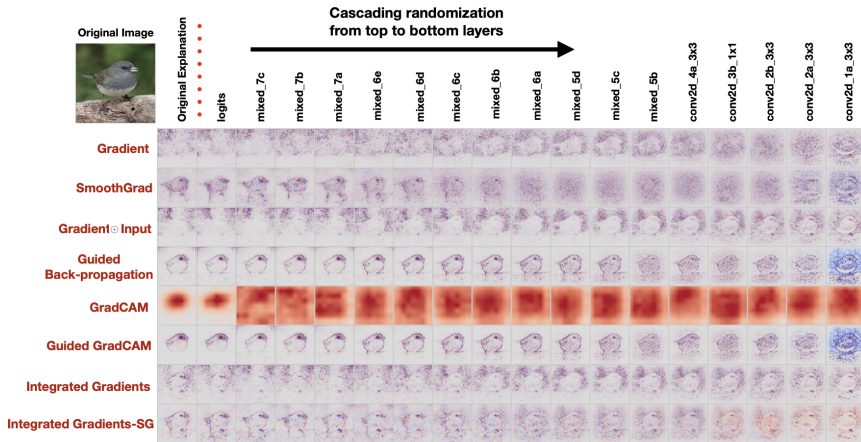
🍷 Sensitivity to Model Parameters ---



If the parameter settings of the model change, the saliency map should change.

- **Model 1**: trained normally
- **Model 2**: randomly initialized (or randomized top layers)
- Saliency maps should **differ** between models

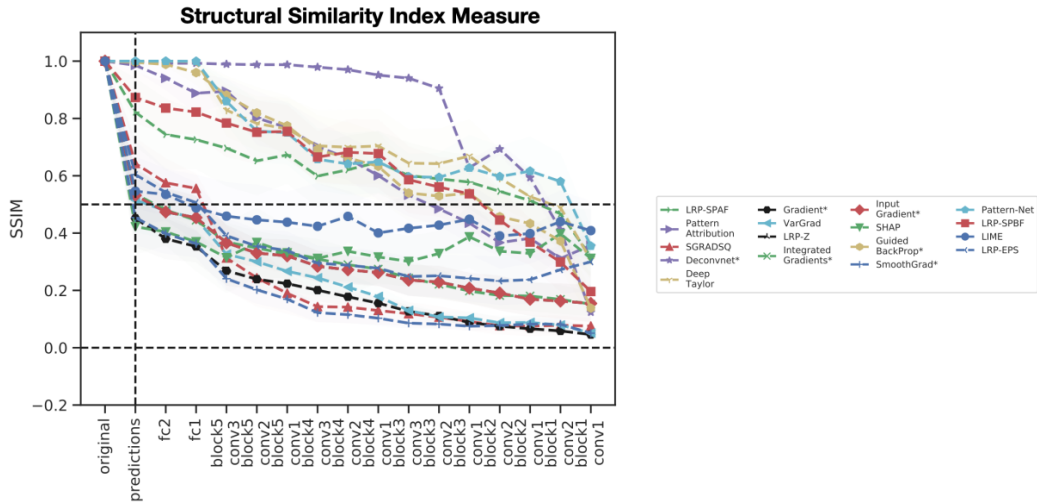
🍷 Sensitivity: Cascading Randomization ---



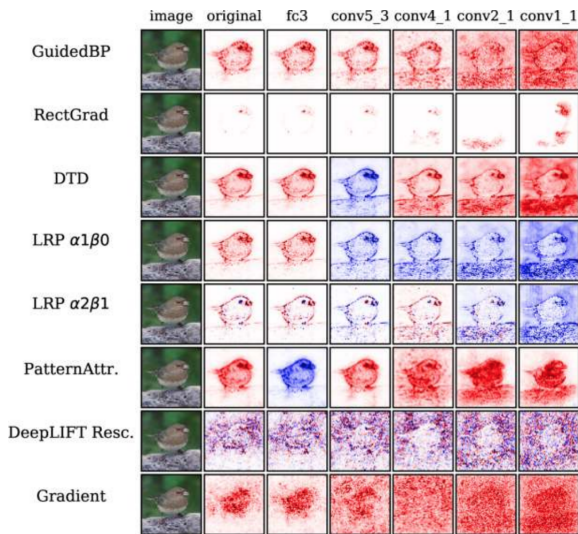
A method that looks the same regardless of the model's parameters cannot be explaining what the model learned.



Sensitivity to Model Parameters: SSIM Results ---



🍷 Modified BackProp Approaches ---



Backprop-based methods that modify gradients at each layer collapse to a **rank-1 matrix** — their output is dominated by the **input structure**, not by the model's learned weights.

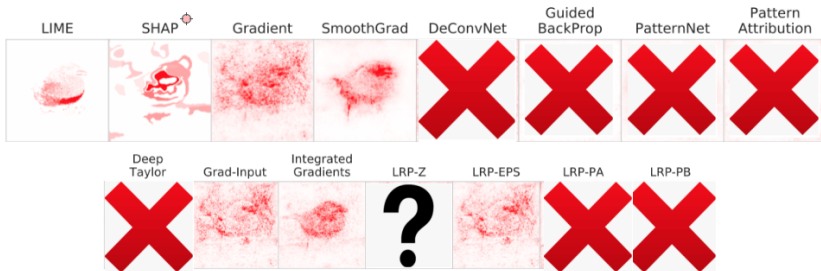
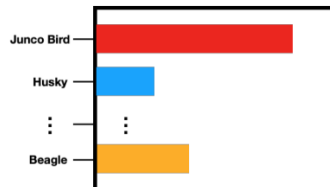
This is why these methods still produce recognizable maps even when the model is completely random: they are recovering the input, not explaining the model.

(Sixt et al. 2020, Theorem 1)

🍷 Recap: Which Method Should You Use? ---



Model



🍷 Some Takeaways ---

- Identified certain classes of feature attribution methods that are **invariant to higher layer weights**
- 'Sanity Checks' are actually **weak** requirements, i.e., does **not** tell you whether a method is effective

Some Objections 1/2 ---

Causal reframing suggests that sanity checks results might be **task specific**.

Revisiting Sanity Checks for Saliency Maps

Gal Yona
Weizmann Institute of Science

Daniel Greenfeld
Jethor Energy Research

On the Relationship Between Explanation and Prediction: A Causal View

Amir-Hossein Karimi^{1 2 3} Krikamol Muandet⁴ Simon Kornblith³ Bernhard Schölkopf¹ Been Kim³

Some Objections 2/2 ---

Where you choose to perform randomization matters, and perhaps the **weight randomization is not the best approach**.

Shortcomings of Top-Down Randomization-Based Sanity Checks for Evaluations of Deep Neural Network Explanations

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Grégoire Montavon^{4,5}[0000-0001-7243-6186], Klaus-Robert Müller^{5,6,7,8}, and Wojciech
Samek^{3,5,6}[0000-0002-6283-3265]

🍲 More Recent Observations: Spurious Features ---

Beyond faithfulness, it is unclear whether these feature attribution methods are effective for **model debugging**.

Do Feature Attribution Methods Correctly Attribute Features?

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🍷 More Recent Observations ---

Do Input Gradients Highlight Discriminative Features?

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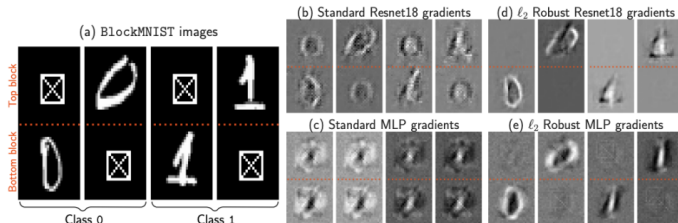
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RETHINKING THE ROLE OF GRADIENT-BASED ATTRIBUTION METHODS FOR MODEL INTERPRETABILITY

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- **BlockMNIST**: each image has two blocks — the discriminative one is fixed by class position. A good saliency method should focus on the correct block.
- **Key finding**: saliency quality depends not only on the attribution method, but also on **how the model was trained** — robustly trained models produce cleaner, more discriminative gradients.

🍷 Parting Thoughts ---

Feature attribution is still important for applications, however, additional work is needed to characterize the properties of DeepNN model training that will result in 'gradients' that capture discriminative signals.

🍷 Exploring the Explanation Landscape ---



OpenXAI: Towards a Transparent Evaluation of Model Explanations

Authors: Chirag Agarwal et al.

(Agarwal et al. 2022)

Overview ---

- Reliability pillars
- OpenXAI
- Is OpenXAI all you need?
- New directions

🍷 Applications of Saliency Methods ---



🍷 Reliability Pillars ---



Pillar 1: Faithfulness ---

Definition

An explanation is faithful if it accurately captures which features the model uses to make its prediction.

$$\text{faithfulness}(e, f, x) = \text{corr}(e_i(x), \Delta f_i(x))$$

where $\Delta f_i(x)$ measures the change in model output when feature i is masked.

🍷 Pillar 2: Stability ---

Definition

An explanation is stable (robust) if similar inputs receive similar explanations — small perturbations to x should not drastically change $e(x)$.



C. Agarwal et al., Rethinking Stability for Attribution-based Explanations, Oral presentation @ ICLR 2022

🍷 Pillar 3: Counterfactual Fairness ---

Definition

An explanation is counterfactually fair if members of different demographic groups receive explanations of comparable quality.

$$\text{PGU} = \mathbb{E}_{x \sim \text{majority}}[\Delta f(x, e)] - \mathbb{E}_{x \sim \text{minority}}[\Delta f(x, e)]$$

J. Dai et al., Fairness via Explanation Quality: Evaluating Disparities in the Quality of Post hoc Explanations, AIES 2022.

How do we **pick** an explanation method
from the XAI landscape?

- OpenXAI provides an **automated end-to-end pipeline** that simplifies and standardizes the evaluation of post hoc explanation methods
- OpenXAI promotes **transparency and reproducibility** in benchmarking explanation methods

C. Agarwal et al., OpenXAI: Towards a Transparent Evaluation of Model Explanations, NeurIPS Datasets and Benchmark Track'2022

<https://github.com/AI4LIFE-GROUP/OpenXAI>

OpenXAI's Key Components ---

- A flexible **synthetic data generator** and a collection of 7 real-world datasets, **16 pre-trained models**, and **6 state-of-the-art explanation methods**
- Open-source implementations of **22 quantitative metrics** for evaluating faithfulness, stability (robustness), and fairness of explanation methods
- **First-ever public XAI leaderboards** to benchmark explanation methods

XAI Ready Dataloaders and Models ---

```
from openxai import Dataloader
loader_train, loader_test = Dataloader.return_loaders(
    data_name='german', download=True)
inputs, labels = iter(loader_test).next()
```

OpenXAI provides pre-trained models for readily benchmarking explanation methods.

```
from openxai import LoadModel
model = LoadModel(data_name='german', ml_model='ann')
```

OpenXAI Explainers ---

OpenXAI provides ready-to-use implementations of six state-of-the-art feature attribution methods.

```
from openxai import Explainer
exp_method = Explainer(method='LIME')
explanations = exp_method.get_explanations(
    model, X=inputs, y=labels)
```

@abstractmethod

```
def get_explanations(self, model, X: torch.Tensor,
                     y: torch.Tensor):
```

```
    """
```

Generate explanations for given input/s.

Parameters: model, X (m x n tensor), y (labels)

Returns: torch.Tensor (explanation vector/matrix)

```
    """
```

OpenXAI's Evaluation ---

OpenXAI provides implementations and ready-to-use APIs for a set of **22 quantitative metrics** proposed by prior research to evaluate the faithfulness, stability, and fairness of explanation methods.

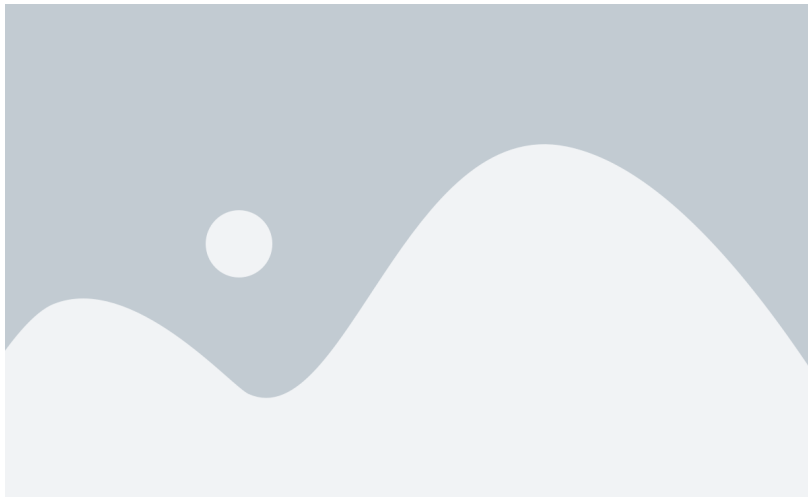
```
from openxai import Evaluator
metric_evaluator = Evaluator(inputs, labels, model,
                             explanations)
score = metric_evaluator.eval(metric='RIS')
```


OpenXAI's Leaderboard ---



🍷 Exploring the Landscape Using OpenXAI ---

- LIME produces more faithful (+24.9%) explanations
- Across all real-world datasets, SmoothGrad achieves **63.2% higher RRS** values



🍷 Is OpenXAI All You Need? ---

- How to benchmark different **non-perturbation-based** explanation methods?
- Benchmarking explanations on **other modalities**:
 - ▶ Vision (Quantus)
 - ▶ NLP (e-ViL)
 - ▶ Graphs (GraphXAI)

C. Agarwal et al., Evaluating Explainability for Graph Neural Networks, Nature Scientific Data'2023

M. Kayser et al., e-ViL: A Dataset and Benchmark for Natural Language Explanations in Vision-Language Tasks, ICCV'2021

New Directions ---

- Training models using **Explanation Feedbacks**
- **Differentiable Explainable Curricula** for RL Agents
- Learning **Hierarchical and Multi-modal Explanations**

Thank You!

References --- |

- Adebayo, Julius, Justin Gilmer, Michael Muelly, Ian Goodfellow, Moritz Hardt, and Been Kim. 2018. "Sanity Checks for Saliency Maps." In *Advances in Neural Information Processing Systems*.
- Agarwal, Chirag, Daniyar Saxena, Satyapriya Krishna, Martin Pawelczyk, Nari Johnson, Isha Puri, Marinka Zitnik, and Himabindu Lakkaraju. 2022. "OpenXAI: Towards a Transparent Evaluation of Model Explanations." In *Advances in Neural Information Processing Systems Datasets and Benchmark Track*.
- Baehrens, David, Timon Schroeter, Stefan Harmeling, Motoaki Kawanabe, Katja Hansen, and Klaus-Robert Müller. 2010. "How to Explain Individual Classification Decisions." In *Journal of Machine Learning Research*.
- Simonyan, Karen, Andrea Vedaldi, and Andrew Zisserman. 2014. "Deep Inside Convolutional Networks: Visualising Image Classification Models and Saliency Maps." In *International Conference on Learning Representations Workshop*.
- Smilkov, Daniel, Nikhil Thorat, Been Kim, Fernanda Viégas, and Martin Wattenberg. 2017. "SmoothGrad: Removing Noise by Adding Noise." In *arXiv Preprint arXiv:1706.03825*.
- Sundararajan, Mukund, Ankur Taly, and Qiqi Yan. 2017. "Axiomatic Attribution for Deep Networks." In *International Conference on Machine Learning*.